

4ai	amplitude is decreasing (very) gradually / oscillations would continue (for a long time) /many oscillations light damping
aii1	$\omega = 2\pi/T$ $= 2\pi/0.3$ $= 20.9 \text{ rad s}^{-1}$
aii2	Loss of Energy = total initial energy of oscillation – total final energy of oscillation = k.e. max initial (when oscillation has an amplitude of 1.5 cm) – k.e. max final (when oscillation has an amplitude of 1.1 cm) $= \frac{1}{2} m \omega^2 x_{o, \text{initial}}^2 - \frac{1}{2} m \omega^2 x_{o, \text{final}}^2$ $= \frac{1}{2} \times 0.065 \times 20.9^2 \times (0.015^2 - 0.011^2)$ $= 0.00147$ $= 0.0015 \text{ J}$
aii3	amplitude <u>reduces exponentially</u> / <u>does not decrease linearly</u> so will be not be 0.7 cm
b	Construction of horizontal line at $E_k = 1.5 \text{ mJ}$ amplitude = 1.1 cm If shifted curve is shown with the correct 3 key points, 2 marks. If shifted curve is shown with only the correct max point (wrong amplitude values), 1 mark

5a	Current = $E / (R+r)$
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